
TOWARDS REVERSIBLE MODEL MERGING FOR LOW-RANK WEIGHTS

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ABSTRACT

Model merging seeks to consolidate multiple fine-tuned models into a single parameter set that attains high performance across all source tasks. While prior work establishes that merging can approximate the per-task accuracy of individually fine-tuned models, it largely neglects the regime in which models are represented in low-rank form—obtained either through low-rank adaptation (LoRA) or through post-training singular value decomposition (SVD). We first establish empirically that applying conventional merging operators to low-rank weight factors induces severe degradation of merged-model accuracy. Motivated by this failure mode, we adopt a substantively different formulation: rather than collapsing all adapters into one parameter set, we construct a compact basis (equivalent in storage footprint to retaining two or more models) from which each original task-specific model can be recovered by linear combination. This recasts merging as the construction of a reconstruction-capable model space rather than the production of a single merged model. Critically, the formulation permits exact “reversion” to any individual model on demand, reflecting the premise that no merged model can uniformly dominate one specialized for its own task. Building on this premise, we introduce Reversible Model Merging (RMM), an efficient, data-free, and configurable procedure that admits a closed-form solution for the optimal weight basis and the task-specific linear-combination coefficients. Extensive experiments spanning diverse datasets and model scales show that RMM consistently outperforms existing merging methods, preserving the accuracy of low-rank compressed models by a substantial margin.

1 INTRODUCTION

Parameter-efficient fine-tuning methods, most prominently Low-Rank Adapters (LoRA) (Hu et al., 2021), have become a standard mechanism for adapting large language models to heterogeneous downstream tasks while avoiding the full computational cost of retraining. In parallel, post-training compression techniques such as truncated singular value decomposition (SVD) are widely used to reduce storage and inference cost, providing an orthogonal axis of efficiency (Ryu et al., 2023; Ping et al., 2024; Wang et al., 2025b;a). Complementing these developments, model merging has emerged as a paradigm for consolidating the knowledge encoded in multiple task-specific models into a single unified parameter set. Recent merging techniques—spanning direct parameter interpolation through optimization-based coefficient search—demonstrate that a merged model can attain competitive accuracy across source tasks without access to the original training data (Matena & Raffel, 2022; Yang et al., 2024; Yadav et al., 2023; Wei et al., 2025).

Despite this progress, a fundamental limitation remains insufficiently addressed: even sophisticated merging strategies systematically fail to recover the accuracy of the original per-task fine-tuned models. This accuracy gap is amplified when the models to be merged are represented in low-rank form, whether through LoRA fine-tuning or through post-training SVD truncation. Although merging yields a single model capable of serving multiple tasks, the operation necessarily projects distinct task-specific representations into one shared parameter space. This projection induces cross-task interference and degrades the specialized adaptations acquired by each model (Gargiulo et al., 2025).

Although each individual low-rank adapter or SVD-truncated update is compact, the accumulation of many such compressed models constitutes a distinct scalability bottleneck. In realistic multi-task, federated, and continual-learning deployments, one routinely accumulates tens to hundreds of task-specific low-rank adapters, each incurring separate storage and coordination overhead. Retaining the full collection erodes the storage savings that low-rank parameterization is intended to deliver and complicates deployment. This motivates merging operators that act directly on low-rank representations—not merely to reduce aggregate storage and redundancy, but to enable more efficient sharing, deployment, and cross-task transfer of compressed models.

To characterize the limitations of current approaches, we first analyze the behavior of existing merging operators applied to low-rank compressed models. The analysis is framed in terms of deltas, defined as the weight differences between each fine-tuned model and the shared pre-trained initialization. Conventional merging strategies aggregate these deltas into a single parameter set under an implicit assumption of linear compatibility. We show empirically, however, that directly merging low-rank deltas produces catastrophic accuracy degradation. We attribute this failure to two mechanisms: (i) the restricted expressive capacity intrinsic to low-rank representations, and (ii) the misalignment of task-specific subspaces, which—being optimized independently—interfere destructively upon aggregation.

Motivated by this observation, we relax the defining constraint of the merging problem and pose three questions: *Must the output of merging be a single final model? Can the problem be relaxed to permit retaining more than one model? Can the ability to reconstruct each original model on demand be preserved?* Rather than collapsing the low-rank adapters into a single weight tensor, we propose to retain a compact set of weights that serves as a shared basis from which each original model is recoverable by linear combination. This reformulation exposes an underexplored design space: instead of compressing all information into one model, we retain complementary representational directions across a small set of basis models, thereby enabling post-merging “reversion” to the original adapters. The approach abandons the prevailing assumption that merging must terminate in a single model and instead defines merging as the construction of a reconstruction-capable model space.

Building on this reconstruction-driven perspective, we develop a theoretical framework that characterizes the basis minimizing reconstruction error. The framework yields our method, Reversible Model Merging (RMM), which reformulates merging as a basis-selection problem: the objective is to identify a compact set of task vectors that supports accurate reconstruction of all original models. Within this formulation, RMM admits a closed-form solution that jointly determines an optimal set of basis task vectors and the task-specific linear-combination coefficients required to recover each individual task adaptation. We validate RMM empirically across a broad range of datasets and model configurations and observe that it consistently surpasses conventional merging strategies by a substantial margin. RMM attains this while retaining the efficiency of low-rank parameterization, which underscores its practicality for large-scale multi-task adaptation. Finally, it exposes a controllable storage–performance trade-off through a single tunable hyperparameter governing the number of retained components.

2 RELATED WORK

Model merging has become an effective strategy for consolidating knowledge across multiple fine-tuned models. Existing methods partition into two categories: data-free parameter-space manipulations, and optimization-based procedures that exploit data or auxiliary objectives. We review representative techniques from each category below.

Task Arithmetic: A foundational formulation in model merging is task arithmetic (Ilharco et al., 2023), in which a delta is defined as the difference between the parameters of a fine-tuned model and its pre-trained initialization. The central claim is that these deltas behave as modular, composable operators in weight space: they may be added to a base model to introduce a capability, subtracted to suppress an undesired behavior, or scaled and summed to blend multiple capabilities. This linear-composition framework provides a principled basis for manipulating model behaviors directly at the parameter level.

DARE: While direct delta addition can be effective, it degrades when the merged models carry conflicting or overlapping parameter updates. To mitigate this, Yu et al. (2024) proposed DARE (Drop and Rescale), which applies aggressive randomized sparsification. The method discards the large majority of parameter changes—a drop rate of 90–99%—under the hypothesis that most updates are redundant and may interfere with knowledge from other models. The surviving updates are then rescaled to preserve the expected magnitude of the modification, retaining the core task-specific updates while suppressing interference.

TIES: In contrast to the randomized sparsification of DARE, TIES-merging specifies a deterministic procedure for resolving parameter conflicts (Yadav et al., 2023). The procedure comprises three stages—Trim, Elect Sign, and Merge—each targeting a specific source of interference. The Trim stage retains only the largest-magnitude parameters, preserving the most influential fine-tuning updates. The Elect Sign stage computes, at each parameter position, the sign agreement of updates across models and discards contributions whose sign opposes the elected majority. The Merge stage averages the sign-consistent contributions to form the merged parameter. This consensus procedure reduces destructive interference and produces a more stable merged model. Following the same interference-reduction objective, EMR-Merging (Elect, Mask, and Rescale) elects a unified model and attaches per-task binary masks and rescaling factors derived from each individual model, steering the unified model toward each task at inference (Huang et al., 2024).

Other Merging Methods: Beyond purely parameter-space manipulations, a second class of merging methods invokes explicit optimization or requires access to a small dataset. Model soups (Wortsman et al., 2022) optimize task-arithmetic interpolation coefficients by minimizing validation loss, steering the interpolation toward improved generalization. Fisher-weighted averaging (Matena & Raffel, 2022) incorporates second-order (Fisher) information, assigning each parameter an importance weight proportional to its estimated criticality for task performance. Other methods cast merging as an explicit optimization problem: RegMean (Jin et al., 2023) and AdaMerging (Yang et al., 2024) search for task-dependent merging coefficients by solving small-scale objectives—respectively, the prediction discrepancy between the merged and individual models, and the entropy of predictions on unlabeled data. A recent formulation minimizes the loss gap between the merged model and the individual fine-tuned models, deriving a tractable surrogate via a first-order Taylor expansion (Wei et al., 2025). Although these methods can improve task-specific accuracy, they require labeled or unlabeled data, additional optimization, or both, which constrains their applicability in settings where merging must proceed without retraining and without data.

3 PRELIMINARIES

We formalize the problem of consolidating a collection of fine-tuned models into a single unified model via merging. Let $\{\theta_1, \dots, \theta_n\}$ denote n models fine-tuned on n distinct tasks. A merging operator $\mathcal{M}$ maps this collection to a single parameter set $\theta_{\mathcal{M}} = \mathcal{M}(\{\theta_i\}_{i=1}^n)$. The objective is to determine a merging operator $\mathcal{M}$ that maximizes the mean validation performance across all tasks:

$$\max_{\mathcal{M}} \frac{1}{n} \sum_{i=1}^n \text{Perf}(\theta_{\mathcal{M}}, \mathcal{D}_i), \quad (1)$$

where $\mathcal{D}_i$ denotes a held-out validation set for task i and Perf denotes an arbitrary performance metric (e.g., accuracy or F_1). Following standard practice, we assume that all fine-tuned models share a common pretrained initialization θ_{pre} . For any layer l , the update (delta) of model i relative to the base model is defined as $\Delta_i^l = \theta_i^l - \theta_{\text{pre}}^l$, where $\Delta_i^l \in \mathbb{R}^{m \times d}$ is the weight difference for a layer of dimension $m \times d$.

3.1 LOW-RANK COMPRESSED MODELS

Fine-tuned models are commonly compressed into low-rank form to reduce storage, training, and inference cost. We consider two widely adopted compression strategies:

LoRA fine-tuning: each layer is augmented with trainable low-rank factors $\mathbf{A} \in \mathbb{R}^{m \times r}$ and $\mathbf{B} \in \mathbb{R}^{r \times d}$, with $r \ll \min(m, d)$. The pretrained weights are held frozen and only $\mathbf{A}$ and $\mathbf{B}$ are optimized during training.

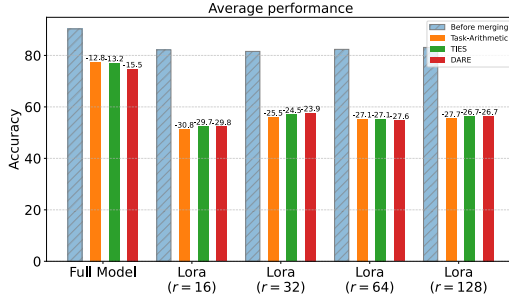


Figure 1: Mean task performance of individual (pre-merge) models against merged models across merging operators. Results are reported for fully fine-tuned models and for LoRA fine-tuning at compression ranks $r \in \{16, 32, 64, 128\}$.

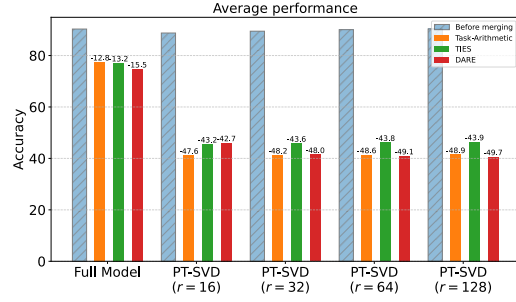


Figure 2: Mean task performance of individual (pre-merge) models against merged models across merging operators. Results are reported for fully fine-tuned models and for PT-SVD compression at compression ranks $r \in \{16, 32, 64, 128\}$.

Post-training SVD truncation (PT-SVD): after standard full fine-tuning, the update Δ_i^l is factorized by SVD and truncated to its top- r singular components, yielding the rank- r approximation $\hat{\Delta}_i^l = A_i^l B_i^l$ with $A_i^l \in \mathbb{R}^{m \times r}$, $B_i^l \in \mathbb{R}^{r \times d}$, and $r \ll \min(m, d)$.

In both cases the fine-tuned update for layer l of model i admits the low-rank factorization $\hat{\Delta}_i^l = A_i^l B_i^l$, with $A_i^l \in \mathbb{R}^{m \times r}$, $B_i^l \in \mathbb{R}^{r \times d}$, and r the compression rank. The merging problem is therefore reduced to combining low-rank updates across tasks. The objective is to construct a merged model that integrates the compressed deltas $\hat{\Delta}_i = \{\hat{\Delta}_i^l\}_{l=1}^L$ over all tasks:

$$\hat{\theta}_{\mathcal{M}} = \theta_{\text{pre}} + \mathcal{M}(\{\hat{\Delta}_i\}_{i=1}^n), \quad \max_{\mathcal{M}} \frac{1}{n} \sum_{i=1}^n \text{Perf}(\hat{\theta}_{\mathcal{M}}, \mathcal{D}_i), \quad (2)$$

where L denotes the total number of layers. Throughout this work we operate in this low-rank regime, in which every model is compressed to low-rank form. The next section characterizes the behavior of conventional merging strategies under this regime.

4 MOTIVATING OBSERVATION

We now characterize the behavior of conventional merging operators applied to the low-rank compressed models introduced in the previous section. For each layer l , prior work considers two natural extensions of merging to the factorized setting (Tang et al., 2025; Zhao et al., 2025; Zheng et al., 2025; Zhang et al., 2025):

- **Combined merging:**

$$\hat{\theta}_{\mathcal{M}}^l = \theta_{\text{pre}}^l + \mathcal{M}(\{A_i^l B_i^l\}_{i=1}^n),$$

in which each low-rank update is first expanded to its full-rank product $A_i^l B_i^l$ prior to merging.

- **Separate merging:**

$$\hat{\theta}_{\mathcal{M}}^l = \theta_{\text{pre}}^l + \mathcal{M}(\{A_i^l\}_{i=1}^n) \mathcal{M}(\{B_i^l\}_{i=1}^n),$$

in which the two low-rank factors are merged independently and the merged factors are subsequently recomposed by multiplication.

Although both formulations are mathematically valid, the combined strategy is prohibitively inefficient. Storing the merged update in combined form requires materializing a dense $m \times d$ matrix, whereas the separate strategy stores only the low-rank factors $A_i^l \in \mathbb{R}^{m \times r}$ and $B_i^l \in \mathbb{R}^{r \times d}$. Since $r(m+d) \ll md$ whenever $r \ll \min(m, d)$, the memory footprint of the combined approach negates the very purpose of compression. Concretely, for a layer with $m = d = 1024$ and compression rank $r = 16$, the combined form stores $md = 1,048,576$ parameters, while the separate form stores $r(m+d) = 16 \times (1024 + 1024) = 32,768$ parameters.

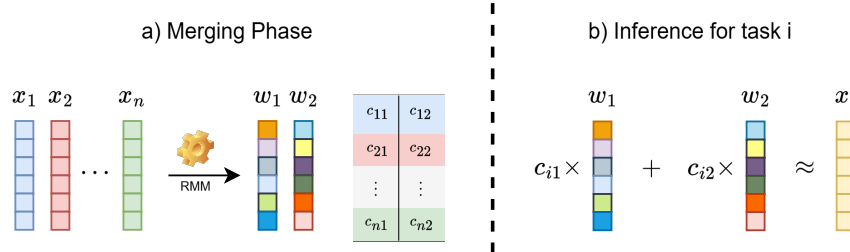


Figure 3: Overview of the RMM framework. **Left:** in the merging phase, RMM computes a compact set of basis directions (here, two) together with their task-specific coefficients. **Right:** at inference, target model i is reconstructed as the linear combination of the basis vectors weighted by its corresponding coefficients.

This corresponds to a $32 \times$ **reduction in storage**, confirming that the combined form eliminates the efficiency gains of low-rank compression. We therefore adopt the separate strategy as the only practically viable operator for merging compressed models.

We evaluated several established merging baselines on low-rank compressed models across a range of compression ranks, using the separate operator defined above. We instantiated the pre-trained model as RoBERTa-base (Zhuang et al., 2021) and evaluated on five GLUE tasks (Wang et al., 2018) (full protocol in Appendix A.3). As shown in Figures 1 and 2, merging compressed models produces a *catastrophic reduction* in mean task performance relative to merging full-rank models. We attribute this degradation to two mechanisms: (i) **Amplified task interference**: compression discards a substantial fraction of the representational capacity, rendering low-rank models more susceptible to destructive interference under aggregation; (ii) **Loss of coupling between A and B** : conventional merging treats the factors A_i^l and B_i^l as independent operands, disregarding that only their product $A_i^l B_i^l$ is semantically meaningful, which induces structural misalignment in the merged update. Jointly, these mechanisms severely degrade merged low-rank models and indicate a fundamental incompatibility between conventional merging operators and compressed representations.

This observation motivates a central conclusion: under conventional strategies, no single merged model can reliably recover the task-specific performance of the original low-rank adapters. This raises a natural question: *can we instead reconstruct, or closely approximate, each original compressed model by retaining a compact set of basis models—equivalent in footprint to two or more adapters, subject to the storage budget?* Rather than forcing all updates into a single representation, we investigate an alternative design space in which merging constructs a *reconstruction-capable model space*. We now formalize this perspective and introduce our method, *Reversible Model Merging* (RMM).

5 REVERSIBLE MODEL MERGING

The empirical failures of conventional merging in the compressed regime expose a fundamental limitation: no single merged model can faithfully preserve the task-specific performance of all low-rank compressed models simultaneously. Motivated by this, RMM abandons the objective of collapsing all models into a single parameter set. Instead, it constructs a small set of shared components (basis models) from which every original task model can be reconstructed to controllable fidelity.

Rather than forcing all n task-specific models into one merged solution $\hat{\theta}_{\mathcal{M}}$, RMM retains a compact set of p merged components, with $2 \leq p \ll n$. These components span a low-dimensional subspace that captures the dominant variation across tasks, permitting efficient on-demand recovery of each model. The number of components p directly parameterizes the performance–memory trade-off: increasing p raises storage cost while decreasing reconstruction error.

For each layer l , we operate on the r -dimensional vectors contained in the low-rank factors A_i^l and B_i^l —specifically, the rows of A_i^l and the columns of B_i^l (recall that r is the compression rank). We designate each such row of A_i^l and each column of B_i^l as an independent *task vector*, and we

term a specific row or column index a *task vector position*. For any fixed position, we aggregate the corresponding task vectors across all n models and denote the task vector contributed by model i at that position as $\mathbf{x}_i \in \mathbb{R}^r$. The algorithm is applied independently at each position: the set of task vectors $\{\mathbf{x}_i\}_{i=1}^n$ is compressed by projection onto a shared basis, from which each vector can subsequently be reconstructed. Each layer contributes $m + d$ task vector positions in total— m rows from $\mathbf{A}_i^l$ and d columns from $\mathbf{B}_i^l$ —and the procedure is executed separately at each.

To this end, we seek a set of basis vectors $\{\mathbf{w}_1, \dots, \mathbf{w}_p\}$ with each $\mathbf{w}_j \in \mathbb{R}^r$, $1 \leq j \leq p$. Their span defines a subspace, represented by the matrix $\mathbf{W} = [\mathbf{w}_1 \dots \mathbf{w}_p] \in \mathbb{R}^{r \times p}$, onto which each task vector $\mathbf{x}_i$ is projected so as to minimize its reconstruction error. We introduce the coefficient matrix $\mathbf{C} \in \mathbb{R}^{n \times p}$, whose i -th row $[c_{i1}, \dots, c_{ip}]$ holds the scalar weights of the linear combination of basis vectors that reconstructs $\mathbf{x}_i$. The objective is to determine $\mathbf{W}$ and $\mathbf{C}$ that jointly minimize the total reconstruction error over the n task vectors, yielding the optimization problem:

$$\min_{\mathbf{W}, \mathbf{C}} \sum_{i=1}^n \left\| \mathbf{x}_i - \sum_{j=1}^p c_{ij} \mathbf{w}_j \right\|_2^2. \quad (3)$$

This representation stores rp parameters for the basis and np for the coefficients, totaling $p(r+n)$ —a substantial reduction from the rn parameters needed to retain all task vectors directly, provided $p \ll \min(r, n)$. The component count p is a tunable hyperparameter governing the performance-storage trade-off. Figure 3 depicts the construction for $p = 2$. The objective in equation 3 can be expressed in matrix form as:

$$\min_{\mathbf{W}, \mathbf{C}} \left\| \mathbf{X} - \mathbf{C} \mathbf{W}^\top \right\|_F^2, \quad (4)$$

where $\mathbf{X} = [\mathbf{x}_1 \dots \mathbf{x}_n]^\top \in \mathbb{R}^{n \times r}$ stacks all task vectors row-wise. For any fixed $\mathbf{W}$, the optimal coefficient matrix is given in closed form by the least-squares solution $\mathbf{C}^* = \mathbf{X} \mathbf{W} (\mathbf{W}^\top \mathbf{W})^{-1}$. Imposing orthonormality on the basis ($\mathbf{W}^\top \mathbf{W} = \mathbf{I}_p$ with $p < r$) reduces the problem to:

$$\min_{\mathbf{W}} \left\| \mathbf{X} - \mathbf{X} \mathbf{W} \mathbf{W}^\top \right\|_F^2 \quad \text{s.t.} \quad \mathbf{W}^\top \mathbf{W} = \mathbf{I}_p. \quad (5)$$

Under this formulation, the stored quantities are $\mathbf{W} \in \mathbb{R}^{r \times p}$ and $\mathbf{X} \mathbf{W} \in \mathbb{R}^{n \times p}$, where $\mathbf{X} \mathbf{W}$ coincides with the coefficient matrix $\mathbf{C}$ of equation 4.

Theorem 1 (Optimal Solution to equation 5). *Let $\mathbf{X} \in \mathbb{R}^{n \times r}$ be zero-centered and $p < \min\{r, n\}$. The optimal basis $\mathbf{W}^*$ that minimizes equation 5 is given by the top- p eigenvectors of the sample covariance matrix $\mathbf{X}^\top \mathbf{X}$. Equivalently, $\mathbf{W}^*$ consists of the top- p right singular vectors of $\mathbf{X}$.*

The proof of Theorem 1 is deferred to Appendix A.1. We now specify the two phases of RMM.

5.1 MERGING PHASE

The merging phase constructs a compact shared-basis representation of all task-specific models. For each layer and each task vector position (a fixed row of $\mathbf{A}_i^l$ or column of $\mathbf{B}_i^l$), we assemble the task vectors from all n models into $\mathbf{X} = [\mathbf{x}_1 \dots \mathbf{x}_n]^\top \in \mathbb{R}^{n \times r}$. We compute the mean $\boldsymbol{\mu} = \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i$ and center the data ($\mathbf{X} \leftarrow [\mathbf{x}_1 - \boldsymbol{\mu} \dots \mathbf{x}_n - \boldsymbol{\mu}]^\top$). We then factorize the centered matrix by SVD, $\mathbf{X} = \mathbf{U} \boldsymbol{\Sigma} \mathbf{V}^\top$, and take its top- p right singular vectors as the optimal orthonormal basis of equation 5, namely $\mathbf{W}^* = \mathbf{V}_{[:,1:p]} \in \mathbb{R}^{r \times p}$. The coefficient matrix is obtained by projection, $\mathbf{C}^* = \mathbf{X} \mathbf{W}^* \in \mathbb{R}^{n \times p}$; its i -th row $[c_{i1}, \dots, c_{ip}]$ holds the linear-combination weights that reconstruct $\mathbf{x}_i$.

On completion of this phase, the original task vectors $\{\mathbf{x}_i\}_{i=1}^n$ are discarded and only the triple $(\mathbf{W}^*, \mathbf{C}^*, \boldsymbol{\mu})$ is retained per task vector position. This reduces the per-position storage from rn parameters (retaining all task vectors) to $p(r+n) + r$. The merging phase is computed offline, prior to any inference.

5.2 INFERENCE PHASE

Consistent with standard model-merging practice, we assume the target task index for each input is either specified explicitly by the user or resolved by an oracle router that assigns the input to the most suitable model. Given the target task index i , the inference phase reconstructs the corresponding task

vector $\mathbf{x}_i$ —denoted $\hat{\mathbf{x}}_i$ —from the compact representation $(\mathbf{W}^*, \mathbf{C}^*, \boldsymbol{\mu})$ retained at each task vector position. The factors $\mathbf{A}_i^l$ and $\mathbf{B}_i^l$ are then recovered by concatenating the reconstructed vectors $\hat{\mathbf{x}}_i$ over all positions (the rows of $\mathbf{A}_i^l$ and the columns of $\mathbf{B}_i^l$, respectively).

Recall that, per task vector position, $\mathbf{W}^* \in \mathbb{R}^{r \times p}$ holds the orthonormal basis, $\mathbf{C}^* \in \mathbb{R}^{n \times p}$ holds the task-specific coefficients, and $\boldsymbol{\mu} \in \mathbb{R}^r$ is the empirical mean over task vectors. Reconstruction proceeds by forming the linear combination of the basis directions with the task-specific coefficients and adding the mean offset $\boldsymbol{\mu}$:

$$\hat{\mathbf{x}}_i = \sum_{j=1}^p c_{ij}^* \mathbf{w}_j^* + \boldsymbol{\mu} = \mathbf{C}_{[i,:]}^* \mathbf{W}^{*\top} + \boldsymbol{\mu}.$$

The resulting $\hat{\mathbf{x}}_i \in \mathbb{R}^r$ approximates the original task vector $\mathbf{x}_i$ discarded during merging. Executing this reconstruction independently at every task vector position (each row of $\mathbf{A}_i^l$ and each column of $\mathbf{B}_i^l$) and across all layers recovers the complete low-rank delta $\hat{\Delta}_i$ for task-specific model i . Algorithm 1 specifies the full merging and inference procedures. We next present an empirical evaluation of RMM against standard baselines.

Algorithm 1 Reversible Model Merging (RMM)

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1: Merging Phase:
2: Require: Low-rank deltas  $\mathbf{A}_i^l$  and  $\mathbf{B}_i^l$ , compression rank  $r$ , number of basis vectors  $p$ 
3: for  $l = 1$  to  $L$  do
4:   for each task vector position (each row in  $\mathbf{A}_i^l$  or each column in  $\mathbf{B}_i^l$ ) do
5:     Gather task vectors  $\mathbf{x}_i$  of each model  $i$  ( $1 \leq i \leq n$ )
6:      $\mathbf{X} \leftarrow [\mathbf{x}_1 \dots \mathbf{x}_n]^\top \in \mathbb{R}^{n \times r}$ 
7:      $\boldsymbol{\mu} \leftarrow \sum_{i=1}^n \mathbf{x}_i \in \mathbb{R}^r$ 
8:      $\mathbf{X} \leftarrow [\mathbf{x}_1 - \boldsymbol{\mu} \dots \mathbf{x}_n - \boldsymbol{\mu}]^\top$ 
9:      $\mathbf{U}, \boldsymbol{\Sigma}, \mathbf{V}^\top \leftarrow \text{SVD}(\mathbf{X})$ 
10:     $\mathbf{W}^* \leftarrow \mathbf{V}_{[:,1:p]} \in \mathbb{R}^{r \times p}$ 
11:     $\mathbf{C}^* \leftarrow \mathbf{X} \mathbf{W}^* \in \mathbb{R}^{n \times p}$ 
12:    Store  $(\mathbf{W}^*, \mathbf{C}^*, \boldsymbol{\mu})$  and discard task vectors  $\{\mathbf{x}_1 \dots \mathbf{x}_n\}$ 
13:   end for
14: end for
15: Inference Phase:
16: Require: target task  $i$ , stored representation  $(\mathbf{W}^*, \mathbf{C}^*, \boldsymbol{\mu})$  for each task vector position
17: for  $l = 1$  to  $L$  do
18:   for each task vector position do
19:      $\hat{\mathbf{x}}_i = \mathbf{C}_{[i,:]}^* \mathbf{W}^{*\top} + \boldsymbol{\mu}$ 
20:   end for
21:    $\hat{\mathbf{A}}_i^l = \text{Concatenate}(\hat{\mathbf{x}}_i \text{ vectors across all row positions})$ 
22:    $\hat{\mathbf{B}}_i^l = \text{Concatenate}(\hat{\mathbf{x}}_i \text{ vectors across all column positions})$ 
23: end for
24:  $\hat{\Delta}_i = \{\hat{\mathbf{A}}_i^l \hat{\mathbf{B}}_i^l\}_{l=1}^L$ 
25: Inference with the reconstructed model  $\theta_{pre} + \hat{\Delta}_i$ 

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6 EXPERIMENTS

To assess the effectiveness of RMM, we conduct extensive experiments spanning multiple compression settings, task suites, and model architectures. The evaluation addresses two questions: whether RMM preserves the accuracy of low-rank task-specific models while reducing aggregate storage cost, and how it compares against conventional merging baselines.

6.1 EXPERIMENTAL SETUP

We first evaluate RMM on eight RoBERTa-base models, each fine-tuned on one of eight distinct tasks (Zhuang et al., 2021). Every model is independently compressed by either PT-SVD or LoRA

r	Method	Storage	QNLI	MRPC	SST-2	MNLI	QQP	RTE	COLA	STS-B	Average
16	No merging	100%	92.02	85.68	93.69	85.21	87.19	77.98	58.05	90.59	83.80
	TA	13%	50.59	33.51	49.08	31.82	36.82	47.29	0.00	4.19	31.66
	TIES	13%	50.56	34.78	49.08	31.82	36.82	47.29	0.00	3.75	31.76
	DARE	13%	50.80	33.51	49.08	31.82	36.82	47.29	0.00	5.07	31.80
	RMM ($p = 2$)	50%	51.20	78.20	88.30	66.74	86.02	47.29	6.56	33.59	57.24
	RMM ($p = 3$)	69%	88.83	86.26	92.89	79.58	87.18	48.74	33.85	60.46	72.22
32	No merging	100%	92.31	85.86	94.15	86.38	88.62	78.34	59.33	90.68	84.46
	TA	13%	50.59	33.51	49.08	31.82	36.82	47.29	0.00	4.21	31.66
	TIES	13%	50.56	35.19	49.08	31.82	36.82	47.29	0.00	3.87	31.83
	DARE	13%	50.72	33.51	49.08	31.82	36.82	47.29	0.00	3.48	31.59
	RMM ($p = 2$)	44%	50.54	74.32	55.05	62.68	87.07	47.29	0.00	23.18	50.02
	RMM ($p = 3$)	59%	84.66	86.32	92.89	79.70	88.05	47.29	23.57	44.54	68.38
64	No merging	100%	92.51	86.03	94.38	87.18	90.15	78.70	61.11	90.72	85.10
	TA	13%	50.61	33.51	49.08	31.82	36.82	47.29	0.00	4.23	31.67
	TIES	13%	50.58	35.13	49.08	31.82	36.82	47.29	0.00	3.80	31.82
	DARE	13%	50.58	33.51	49.08	31.82	36.82	47.29	0.00	5.22	31.79
	RMM ($p = 2$)	41%	50.53	73.10	49.08	58.48	86.98	47.29	0.00	17.50	47.87
	RMM ($p = 3$)	55%	74.14	86.20	92.43	78.83	89.00	47.29	13.16	33.70	64.34
128	No merging	100%	92.68	86.14	94.50	87.33	91.05	79.78	60.16	90.74	85.30
	TA	13%	50.59	33.51	49.08	31.82	36.82	47.29	0.00	4.23	31.67
	TIES	13%	50.58	35.54	49.08	31.82	36.82	47.29	0.00	3.93	31.88
	DARE	13%	50.61	34.78	49.08	31.82	36.82	47.29	0.00	2.41	31.60
	RMM ($p = 2$)	39%	50.54	70.78	49.08	54.69	84.68	47.29	0.00	14.53	46.45
	RMM ($p = 3$)	52%	59.20	85.80	92.32	77.50	89.02	47.29	6.56	26.94	60.58

Table 1: GLUE performance for merging eight RoBERTa-base models under PT-SVD compression at ranks $r \in \{16, 32, 64, 128\}$. *Storage* denotes the fraction of parameters retained relative to storing all individual adapters.

at target ranks $r \in \{16, 32, 64, 128\}$. RMM is then applied to the resulting low-rank models and compared against three standard merging baselines: Task Arithmetic (TA) (Ilharco et al., 2023), TIES-merging (Yadav et al., 2023), and DARE (Yu et al., 2024). As an upper bound, we report the accuracy of the individual compressed models prior to merging. Appendix A.2 reports additional experiments on OPT-1.3b (Zhang et al., 2022) and on ViT-B/32 (Radford et al., 2021) for vision tasks, under differing task counts.

We adopt the GLUE benchmark protocol (Wang et al., 2018), scoring each task with its standard metric. QNLI (question–answering inference), MRPC (paraphrase detection), SST-2 (sentiment classification), MNLI (multi-genre natural language inference), QQP (duplicate question detection), and RTE (textual entailment) are scored by accuracy; CoLA (linguistic acceptability) by Matthews correlation; and STS-B (semantic textual similarity) by Pearson correlation. We summarize overall performance by the unweighted mean across tasks. We further report relative storage cost, defined as the fraction of parameters retained relative to storing all individual task-specific adapters. Baselines that retain a single merged model incur a fixed cost of $\frac{r}{rn} = \frac{1}{n}$, whereas RMM with p basis components incurs $\frac{p(r+n)+r}{rn}$, as derived in Section 5.1.

6.2 RESULTS AND DISCUSSION

Tables 1 and 2 report RMM against the baselines under PT-SVD and LoRA compression, respectively. Across both compression schemes and all target ranks $r \in \{16, 32, 64, 128\}$, RMM outperforms TA, TIES, and DARE by a large margin. At the lower ranks ($r = 16, 32$), the baselines fail catastrophically, attaining mean scores of only 31–42% and degrading most severely on the structurally sensitive tasks CoLA and STS-B. By contrast, RMM at $p = 2$ yields substantial gains (e.g., 57.24% under PT-SVD and 51.41% under LoRA at $r = 16$), and RMM at $p = 3$ improves further (up to 72.22% and 58.64% at $r = 16$), markedly narrowing the gap to the individual compressed models. At the higher ranks ($r = 64, 128$), where compression error is smaller, RMM continues to dominate: at $p = 3$ it attains 64.34% and 60.58% under PT-SVD and 55.40% and 61.35% under LoRA, uniformly exceeding the single-model baselines, which remain near 40%. Even at $p = 2$, RMM retains a clear advantage, confirming that the reversible formulation avoids the destructive interference that degrades prior methods.

r	Method	Storage	QNLI	MRPC	SST-2	MNLI	QQP	RTE	COLA	STS-B	Average
16	No merging	100%	61.38	83.48	94.50	83.90	87.56	68.23	53.92	88.71	77.71
	TA	13%	50.59	33.51	72.13	34.79	64.46	47.29	0.00	21.81	40.57
	TIES	13%	50.63	33.68	74.31	35.64	67.45	47.29	0.00	26.38	41.92
	DARE	13%	50.63	33.51	73.74	35.97	65.35	47.29	0.00	21.62	41.01
	RMM ($p = 2$)	50%	51.13	34.14	90.71	60.48	84.41	47.29	0.00	43.14	51.41
	RMM ($p = 3$)	69%	55.92	38.14	93.12	82.95	87.30	47.29	6.56	57.81	58.64
32	No merging	100%	60.13	81.28	93.81	84.46	88.18	62.82	53.12	88.53	76.54
	TA	13%	50.91	68.06	54.70	33.77	65.50	55.96	0.00	-0.48	41.05
	TIES	13%	51.03	68.06	56.77	34.24	66.27	58.12	0.00	1.02	41.94
	DARE	13%	50.96	64.64	63.07	35.12	63.66	55.60	0.00	1.01	41.76
	RMM ($p = 2$)	44%	52.08	65.39	81.54	62.46	77.06	57.40	0.00	9.95	50.73
	RMM ($p = 3$)	59%	54.62	69.97	92.20	81.29	85.97	57.40	4.64	24.08	58.77
64	No merging	100%	59.62	84.12	94.15	85.20	88.75	67.15	58.31	88.70	78.25
	TA	13%	54.75	68.52	77.29	37.24	36.82	47.29	0.00	-3.53	39.80
	TIES	13%	54.82	68.75	80.73	37.67	36.82	47.29	0.00	-2.48	40.45
	DARE	13%	53.71	68.58	78.56	38.33	36.82	47.29	0.00	-4.28	39.88
	RMM ($p = 2$)	41%	55.74	69.28	87.27	42.81	77.72	47.29	0.00	4.91	48.13
	RMM ($p = 3$)	55%	59.07	70.14	90.83	79.11	82.06	47.29	0.00	14.70	55.40
128	No merging	100%	61.58	85.04	93.58	85.71	89.68	57.04	59.35	89.68	77.71
	TA	13%	51.49	69.22	54.47	37.16	63.27	47.29	0.00	-1.74	40.15
	TIES	13%	51.91	68.93	57.34	37.81	63.36	47.29	0.00	-0.25	40.80
	DARE	13%	51.97	68.99	58.60	35.64	63.33	47.29	0.00	-0.44	40.67
	RMM ($p = 2$)	39%	57.18	67.83	88.53	49.54	84.64	47.29	0.00	28.00	52.88
	RMM ($p = 3$)	52%	59.13	39.19	92.09	80.08	86.39	47.29	15.45	71.18	61.35

Table 2: GLUE performance for merging eight RoBERTa-base models under LoRA compression at ranks $r \in \{16, 32, 64, 128\}$. *Storage* denotes the fraction of parameters retained relative to storing all individual adapters.

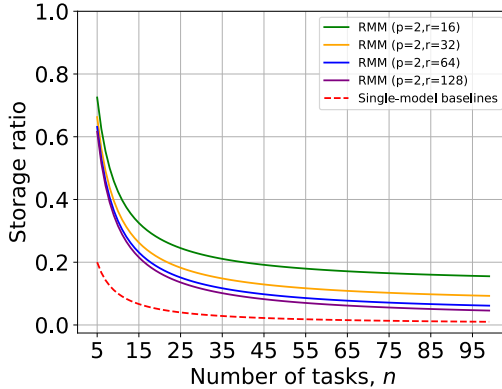


Figure 4: Scalability of RMM ($p = 2$). The storage ratio is the memory required as a percentage of that needed to store all individual models.

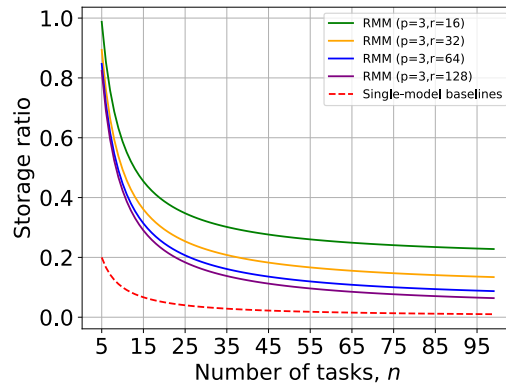


Figure 5: Scalability of RMM ($p = 3$). The storage ratio is the memory required as a percentage of that needed to store all individual models.

We also observe that the relative gain of RMM over the baselines contracts as the rank increases. A plausible explanation is that higher-rank task representations occupy higher-dimensional, more expressive, and less mutually aligned subspaces, which impedes their compression into a shared low-dimensional basis. Overall, the results establish a consistent advantage for RMM across both compression schemes and all ranks: it preserves high task accuracy at the cost of moderate additional storage, and it exposes a tunable accuracy–storage trade-off through the number of reversible components p . Critically, this additional cost is borne by **offline memory**—typically CPU-resident and inexpensive—whereas task accuracy governs runtime quality. Under this asymmetry, the modest increase in CPU-side storage is generally a favorable trade for a substantial accuracy improvement.

6.3 SCALABILITY OF RMM

We next analyze the scaling behavior of RMM as the number of models n grows. Figures 4 and 5 plot the relative storage ratio (with respect to storing all individual task-specific models) as a

function of the task count for $p = 2$ and $p = 3$, respectively. The relative storage cost decreases monotonically as the number of tasks grows, confirming favorable scaling. This behavior reflects the defining property of reversible merging: storage grows sublinearly in the number of task-specific models. The sublinearity follows directly from the RMM construction—once the basis $\mathbf{W}^*$ is fixed, incorporating a new task requires only one additional row in the coefficient matrix $\mathbf{C}^*$ (a cost of p per position). Conventional approaches, by contrast, either retain a single merged model (sacrificing accuracy) or store each task-specific model independently (incurring linear growth). These properties make RMM well suited to large-scale multi-task, continual, and federated settings that must manage hundreds of models efficiently without compromising task accuracy.

7 CONCLUSION

We introduced a principled framework for reversible model merging that reformulates how multiple fine-tuned models are stored, merged, and reconstructed. Rather than compressing all task-specific knowledge into a single merged model, the framework relaxes this constraint by retaining a compact set of intermediate components—substantially fewer than the number of original models, yet strictly more expressive than a single aggregate. This construction supports accurate reconstruction of each task-specific model at a significantly reduced storage cost, and it exposes a controllable accuracy–storage trade-off, particularly in the low-rank compressed regime. By representing fine-tuned models through a compact basis and reversible linear mappings, we demonstrate that models can be merged, reconstructed, and deployed at high accuracy under strict memory budgets, with the compression ratio and reconstruction fidelity tunable to the requirements of a given application.

More broadly, the framework advances beyond heuristic, one-shot merging by formalizing merging as a structured, reversible operation. This formulation both broadens the design space for efficient multi-task deployment and identifies concrete directions for future work in performance-aware compression and reconstruction.

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A APPENDIX

A.1 PROOF OF THEOREM 1

Define the residual $\mathbf{A} = \mathbf{X} - \mathbf{X}\mathbf{W}\mathbf{W}^\top$. By definition of the Frobenius norm, $\|\mathbf{A}\|_F^2 = \text{Tr}(\mathbf{A}^\top \mathbf{A})$, where $\text{Tr}(\cdot)$ denotes the trace of a square matrix. Expanding the squared norm yields:

$$\|\mathbf{X} - \mathbf{X}\mathbf{W}\mathbf{W}^\top\|_F^2 = \text{Tr}((\mathbf{X} - \mathbf{X}\mathbf{W}\mathbf{W}^\top)^\top (\mathbf{X} - \mathbf{X}\mathbf{W}\mathbf{W}^\top)).$$

Expanding the argument of the trace gives:

$$\begin{aligned} (\mathbf{X} - \mathbf{X}\mathbf{W}\mathbf{W}^\top)^\top (\mathbf{X} - \mathbf{X}\mathbf{W}\mathbf{W}^\top) &= (\mathbf{X}^\top - \mathbf{W}\mathbf{W}^\top \mathbf{X}^\top)(\mathbf{X} - \mathbf{X}\mathbf{W}\mathbf{W}^\top) \\ &= \mathbf{X}^\top \mathbf{X} - \mathbf{X}^\top \mathbf{X}\mathbf{W}\mathbf{W}^\top - \mathbf{W}\mathbf{W}^\top \mathbf{X}^\top \mathbf{X} \\ &\quad + \mathbf{W}\mathbf{W}^\top \mathbf{X}^\top \mathbf{X}\mathbf{W}\mathbf{W}^\top. \end{aligned}$$

Taking the trace of the expanded form and applying the cyclic property of the trace together with the orthonormality constraint $\mathbf{W}^\top \mathbf{W} = \mathbf{I}_p$ gives:

$$\begin{aligned} \|\mathbf{X} - \mathbf{X}\mathbf{W}\mathbf{W}^\top\|_F^2 &= \text{Tr}(\mathbf{X}^\top \mathbf{X}) - 2\text{Tr}(\mathbf{W}^\top \mathbf{X}^\top \mathbf{X}\mathbf{W}) + \text{Tr}(\mathbf{W}^\top \mathbf{W}\mathbf{W}^\top \mathbf{X}^\top \mathbf{X}\mathbf{W}) \\ &= \text{Tr}(\mathbf{X}^\top \mathbf{X}) - 2\text{Tr}(\mathbf{W}^\top \mathbf{X}^\top \mathbf{X}\mathbf{W}) + \text{Tr}(\mathbf{W}^\top \mathbf{X}^\top \mathbf{X}\mathbf{W}) \\ &= \text{Tr}(\mathbf{X}^\top \mathbf{X}) - \text{Tr}(\mathbf{W}^\top \mathbf{X}^\top \mathbf{X}\mathbf{W}). \end{aligned}$$

Because $\text{Tr}(\mathbf{X}^\top \mathbf{X})$ is independent of $\mathbf{W}$, minimizing $\|\mathbf{X} - \mathbf{X}\mathbf{W}\mathbf{W}^\top\|_F^2$ is equivalent to maximizing $\text{Tr}(\mathbf{W}^\top \mathbf{X}^\top \mathbf{X}\mathbf{W})$. Therefore,

$$\min_{\mathbf{W}^\top \mathbf{W} = \mathbf{I}_p} \|\mathbf{X} - \mathbf{X}\mathbf{W}\mathbf{W}^\top\|_F^2 \iff \max_{\mathbf{W}^\top \mathbf{W} = \mathbf{I}_p} \text{Tr}(\mathbf{W}^\top \mathbf{X}^\top \mathbf{X}\mathbf{W}).$$

By the Rayleigh quotient theorem (Horn & Johnson, 2012), this trace is maximized by setting the columns of $\mathbf{W}$ to the top- p eigenvectors of $\mathbf{X}^\top \mathbf{X}$. This is exactly the principal component analysis (PCA) solution: it identifies the p -dimensional subspace that maximizes the projected variance and, equivalently, minimizes the squared reconstruction error. Furthermore, given the SVD $\mathbf{X} = \mathbf{U}\Sigma\mathbf{V}^\top$, we have $\mathbf{X}^\top \mathbf{X} = \mathbf{V}\Sigma^2\mathbf{V}^\top$, so the eigenvectors of $\mathbf{X}^\top \mathbf{X}$ are precisely the right singular vectors of $\mathbf{X}$ (the columns of $\mathbf{V}$). The PCA solution is thus equivalently obtained from the top- p right singular vectors of $\mathbf{X}$. In practice, this SVD-based route is numerically more stable and often computationally preferable to explicitly forming $\mathbf{X}^\top \mathbf{X}$, particularly when r is large.

A.2 ADDITIONAL RESULTS

This section provides extended results for two additional pre-trained backbones: OPT-1.3b and ViT-B/32. For **OPT-1.3b**, we evaluate on six natural language understanding tasks drawn from GLUE and related benchmarks: QNLI, MRPC, SST-2, MNLI, QQP, and BoolQ (Clark et al., 2019). For **ViT-B/32**, we evaluate on seven diverse image classification datasets: MNIST (LeCun et al., 1998), GTSRB (Stallkamp et al., 2011), SVHN (Netzer et al., 2011), Cars (Krause et al., 2013), EuroSAT (Helber et al., 2019), DTD (Cimpoi et al., 2014), and SUN397 (Xiao et al., 2010).

Table 3 reports results for six OPT-1.3b models under PT-SVD compression. Both RMM variants ($p = 2$ and $p = 3$) consistently exceed the baselines, and at $p = 3$ RMM narrows the gap to the individual models while retaining substantially fewer parameters. Table 4 reports results for seven ViT-B/32 models. The advantage of RMM is more pronounced in the vision setting: at $p = 3$, mean accuracy is within 3% of the individual models while storage is reduced to 59% of that required to retain all models at $r = 128$. These results confirm that RMM reconstructs the original compressed models with high fidelity while delivering substantial storage efficiency.

A.3 RESULTS FOR 5 TASKS

For the experiments of Section 4, we instantiated the pretrained model as RoBERTa-base and evaluated the merging baselines on five GLUE tasks (QNLI, MRPC, SST-2, MNLI, QQP); Figures 1 and 2 in Section 4 summarize these results. For completeness, we tabulate the underlying numbers: Table 5 corresponds to Figure 1 and reports the LoRA fine-tuned models together with the improvement attained by RMM, and Table 6 corresponds to Figure 2 and reports the PT-SVD compressed models.

r	Method	Storage	QNLI	MRPC	SST-2	MNLI	QQP	BOOLQ	Average
16	No merging	100%	91.74	71.54	95.30	84.19	88.14	71.47	83.73
	Task-Arithmetic	17%	50.34	37.16	58.49	34.43	63.30	41.87	47.60
	TIES	17%	49.79	36.75	61.24	35.57	63.55	40.67	47.93
	DARE	17%	49.95	37.68	56.08	34.51	63.39	42.78	47.40
	RMM ($p = 2$)	63%	88.12	44.00	93.58	73.06	85.63	59.69	74.01
	RMM ($p = 3$)	85%	91.58	49.04	95.64	80.31	87.52	65.50	78.27
32	No merging	100%	92.06	72.17	95.41	84.60	89.17	72.45	84.31
	Task-Arithmetic	17%	50.28	37.22	58.94	34.57	63.33	41.93	47.71
	TIES	17%	49.77	36.93	61.93	35.69	63.65	40.73	48.12
	DARE	17%	50.39	40.87	56.19	34.96	63.59	43.33	48.22
	RMM ($p = 2$)	56%	86.66	41.91	91.74	71.75	85.68	56.39	72.36
	RMM ($p = 3$)	76%	91.85	44.81	94.84	80.21	88.43	63.52	77.28
64	No merging	100%	92.26	72.12	95.41	84.64	90.07	72.84	84.56
	Task-Arithmetic	17%	50.28	37.28	59.06	34.56	63.34	42.08	47.77
	TIES	17%	49.75	37.04	61.81	35.84	63.74	40.76	48.16
	DARE	17%	50.16	40.52	58.83	35.59	63.35	42.54	48.50
	RMM ($p = 2$)	53%	83.87	42.72	89.45	69.76	85.70	53.43	70.82
	RMM ($p = 3$)	71%	92.07	41.97	94.50	79.70	88.73	61.96	76.49
128	No merging	100%	92.35	72.41	95.41	84.67	90.93	73.09	84.81
	Task-Arithmetic	17%	50.32	37.39	58.94	34.63	63.37	42.08	47.79
	TIES	17%	49.79	37.04	62.04	35.99	63.74	40.55	48.19
	DARE	17%	50.03	38.14	57.91	34.72	63.31	42.11	47.70
	RMM ($p = 2$)	52%	81.38	42.78	84.75	67.66	83.22	51.99	68.63
	RMM ($p = 3$)	69%	92.28	41.80	93.69	79.42	88.76	59.69	75.94

Table 3: GLUE performance for merging six OPT-1.3b models under PT-SVD compression at ranks $r \in \{16, 32, 64, 128\}$. *Storage* denotes the fraction of parameters retained relative to storing all individual adapters.

r	Method	Storage	MNIST	GTSRB	SVHN	Cars	EuroSAT	DTD	SUN397	Average
16	No merging	100%	99.61	97.22	97.06	69.20	99.67	75.21	72.72	87.24
	Task-Arithmetic	14%	47.19	28.29	38.64	56.27	57.52	41.38	62.92	47.46
	TIES	14%	60.14	34.52	49.03	58.55	61.70	43.67	62.87	52.93
	DARE	14%	53.80	31.00	44.96	57.60	56.93	43.99	62.42	50.10
	RMM ($p = 2$)	55%	99.16	88.52	94.89	66.61	97.44	60.64	67.86	82.16
	RMM ($p = 3$)	76%	99.58	95.65	96.54	68.30	98.93	67.18	69.50	85.10
32	No merging	100%	99.65	98.45	97.35	73.04	99.78	78.72	75.52	88.93
	Task-Arithmetic	14%	47.90	28.31	38.68	56.24	57.52	41.38	63.00	47.58
	TIES	14%	61.25	35.28	49.95	58.80	62.41	44.04	63.05	53.54
	DARE	14%	53.05	31.04	46.34	58.54	57.74	42.87	62.87	50.35
	RMM ($p = 2$)	49%	98.98	88.23	94.29	68.32	96.81	61.91	68.36	82.41
	RMM ($p = 3$)	67%	99.50	96.53	96.68	70.90	98.93	70.43	71.23	86.31
64	No merging	100%	99.67	98.68	97.41	75.76	99.85	79.15	77.73	89.75
	Task-Arithmetic	14%	47.99	28.30	38.71	56.21	57.59	41.38	63.22	47.63
	TIES	14%	61.83	35.93	50.28	59.08	63.07	44.15	63.17	53.93
	DARE	14%	54.78	32.79	45.05	58.23	61.30	42.87	62.77	51.11
	RMM ($p = 2$)	46%	98.46	84.43	92.83	69.36	95.59	62.39	69.77	81.83
	RMM ($p = 3$)	62%	99.45	96.14	96.33	72.84	98.56	72.07	73.85	87.03
128	No merging	100%	99.68	98.75	97.46	76.99	99.85	79.04	78.59	90.05
	Task-Arithmetic	14%	48.00	28.31	38.73	56.21	57.63	41.44	63.35	47.67
	TIES	14%	62.08	36.36	50.48	59.23	63.41	44.68	63.25	54.21
	DARE	14%	55.16	32.07	45.47	58.55	61.85	43.09	62.85	51.29
	RMM ($p = 2$)	44%	97.22	78.56	90.63	70.22	94.00	61.49	71.03	80.45
	RMM ($p = 3$)	59%	99.28	94.59	95.95	74.24	98.19	71.86	76.02	87.16

Table 4: Vision-task performance for merging seven ViT-B/32 models under PT-SVD compression at ranks $r \in \{16, 32, 64, 128\}$. *Storage* denotes the fraction of parameters retained relative to storing all individual adapters.

r	Method	Storage	QNLI	MRPC	SST-2	MNLI	QQP	Average
16	No merging	100%	61.38	83.48	94.50	83.90	87.56	82.16
	Task-Arithmetic	20%	50.61	33.51	73.85	34.23	64.58	51.36
	TIES	20%	50.65	33.51	75.57	35.83	66.97	52.51
	DARE	20%	50.61	33.51	79.36	34.70	63.40	52.32
	RMM ($p = 2$)	73%	59.64	68.64	93.00	82.68	86.89	78.17
32	No merging	100%	60.13	81.28	93.81	84.46	88.18	81.57
	Task-Arithmetic	20%	51.18	67.30	58.83	34.33	68.56	56.04
	TIES	20%	51.36	68.17	59.52	34.76	71.46	57.05
	DARE	20%	50.83	64.87	68.12	35.54	68.93	57.66
	RMM ($p = 2$)	66%	57.72	67.54	92.55	81.80	86.13	77.15
64	No merging	100%	59.62	84.12	94.15	85.20	88.75	82.37
	Task-Arithmetic	20%	54.74	68.93	80.28	35.70	36.82	55.29
	TIES	20%	55.26	68.93	78.78	36.76	36.82	55.31
	DARE	20%	54.71	68.81	74.54	38.72	36.82	54.72
	RMM ($p = 2$)	63%	59.82	67.48	91.51	83.21	84.41	77.29
128	No merging	100%	61.58	85.04	93.58	85.71	89.68	83.12
	Task-Arithmetic	20%	52.52	68.70	54.82	37.83	63.46	55.47
	TIES	20%	52.55	69.04	57.80	38.91	63.72	56.40
	DARE	20%	55.04	69.62	54.47	38.92	64.12	56.43
	RMM ($p = 2$)	62%	60.21	34.96	91.86	78.59	86.65	70.45

Table 5: GLUE performance for merging five RoBERTa-base models under LoRA compression at ranks $r \in \{16, 32, 64, 128\}$. *Storage* denotes the fraction of parameters retained relative to storing all individual adapters.

r	Method	Storage	QNLI	MRPC	SST-2	MNLI	QQP	Average
16	No merging	100%	92.02	85.68	93.69	85.21	87.19	88.76
	Task-Arithmetic	20%	53.03	35.19	49.08	31.82	36.82	41.19
	TIES	20%	55.46	54.32	49.08	31.82	37.21	45.58
	DARE	20%	53.69	58.67	49.08	31.82	37.08	46.07
	RMM ($p = 2$)	73%	79.72	85.16	90.83	76.58	86.09	83.68
32	No merging	100%	92.31	85.86	94.15	86.38	88.62	89.46
	Task-Arithmetic	20%	53.23	35.59	49.08	31.82	36.82	41.31
	TIES	20%	55.78	55.19	49.08	31.82	37.62	45.90
	DARE	20%	55.89	33.74	49.20	31.82	36.85	41.50
	RMM ($p = 2$)	66%	63.87	82.78	91.28	77.12	87.05	80.42
64	No merging	100%	92.51	86.03	94.38	87.18	90.15	90.05
	Task-Arithmetic	20%	53.49	35.88	49.08	31.82	36.82	41.42
	TIES	20%	56.14	55.94	49.08	31.82	38.08	46.21
	DARE	20%	53.51	33.68	49.08	31.82	36.82	40.98
	RMM ($p = 2$)	63%	56.12	81.22	91.06	76.74	88.07	78.64
128	No merging	100%	92.68	86.14	94.50	87.33	91.05	90.34
	Task-Arithmetic	20%	53.60	36.06	49.08	31.82	36.82	41.48
	TIES	20%	56.40	56.35	49.08	31.82	38.61	46.45
	DARE	20%	50.52	34.96	49.08	31.82	36.92	40.66
	RMM ($p = 2$)	62%	54.11	79.19	90.60	74.49	87.46	77.17

Table 6: GLUE performance for merging five RoBERTa-base models under PT-SVD compression at ranks $r \in \{16, 32, 64, 128\}$. *Storage* denotes the fraction of parameters retained relative to storing all individual adapters.